

**Sudhir Power Ltd.**  
Power Generation Business Unit

# Engine Test Certificate

This is to certify that Engine Model **6BTAA5.9-G13**

Serial Number **P8149078** was tested using

Diesel as per ISO – 3046 Standard.

Corrected Full Load B.H.P. to N.T.P. conditions as per the

Test standard is **159** B.H.P. @ **1800** R.P.M.

Date: 23-FEB-16

  
Product Conformance

Sudhir Gensets Ltd. ATHAL UNIT

**sudhir**

Powering the Nation

Plot No. 8,9 & 10, Survey No. 144/1/2/1/2  
& 144/1/2/1/3 & 144/1/2/10. Athal Industrial Estate, Village - Athal.  
Post. Naroli, Silvassa. (U.T. of D & N.H.) Tel.: (0260) 3291420 / 21

## TEST CERTIFICATE FOR ELECTRIC DIESEL GENERATING SET

This is to certify that 125 KVA Diesel Generating Set

S. R. No. 1404B292 has been tested at our works as per  
ISO : 8528. The performance of the set was found electrically  
and mechanically sound and in working order in all particulars.

The set was coupled with the following major components.

Engine Make :

*Unimac*

Engine Model : *6B7000A9413*

Engine Sr. No. : P8149078

Alternator Make :

*Starfed*

Frame Size : *W4274N1*

Alternator Sr. No. N14C133169

*[Signature]*  
(Authorised Signatory)  
(Q. A & C.)



**STAMFORD****Test Certificate**FOR  
SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATORMachine No. **N14C133169**Type Test has been conducted on similar design machine  
STANDARDS: Generator conforms to BS 5000 : Part 3, IS 4722  
Tested in accordance with BS EN 60034 - 1  
Rotor dynamically balanced to BS ISO 1940 - 1**ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:**

|                  |                       |                    |                         |
|------------------|-----------------------|--------------------|-------------------------|
| FRAME: UCI274V1  | AVR: SX460            | AVR NO: 1072590494 | ENCLOSURE: IP23         |
| RATED KVA: 125   | RATED VOLTS: 400      | RATED AMPS: 173.9  | FREQUENCY: 60           |
| POWER FACTOR: .8 | STR CONN: Series Star | STR WINDING: 311   | RATING: Base Continuous |
| EXC. VOLTS: 60   | EXC. AMPS: 2.5        | AMB TEMP: 40       | PHASE: 3                |
|                  |                       |                    | WIRE: 4-Wire            |

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION : U-V-W

| WINDING | COLD RESISTANCE<br>IN Ohm (+/- 10) AT 20°C<br>(MAX VALUES) | INSULATION<br>RESISTANCE<br>(M- Ohm) | HV TEST 2KV<br>FOR 1 MIN | INSULATION CLASS |
|---------|------------------------------------------------------------|--------------------------------------|--------------------------|------------------|
| STATOR  | 0.08296                                                    | 2+                                   | OK                       | H                |
| ROTOR   | 1.399                                                      | 2+                                   | OK                       | H                |

**REGULATION TEST**

| FREQ. | VOLTS | AMPS  | KVA  | PF   | KW   | EXC. VOLT | EXC. AMP | % LOAD |
|-------|-------|-------|------|------|------|-----------|----------|--------|
| 62    | 401.4 | 0.00  | 0.00 | 0.00 | 0.00 | 11        | .46      | 0.00   |
| 60    | 400   | 173.9 | 125  | .8   | 100  | 60        | 2.5      | 100.00 |

REGULATION = (N.L. VOLTAGE - F.L. VOLTAGE / RATED VOLTAGE) \* 100 = 0.34 %

Item Description: AC Generator, UC27.4.V.1.311., 3., 11.5., SX460. Onan Green  
Remarks:

Date: 23-FEB-16

Job No: 3366557-009

AUTHORISED SIGNATORY

Quality Assurance

**Certificate Of Conformity**Quality System Approval  
ISO 9001/2008Frame Size : **UCI274V1**Serial Number : **N14C133169**

Manufactured Date : 23-FEB-16

KVA: **125**Authorised Signatory Quality Assurance: 

Electrical Test

Final Inspection

T01

T07

Date: 23-FEB-16